A Controlled Reasoning-SFT Trajectory Reveals Multi-Timescale Routing Reorganization in Sparse Mixture-of-Experts Models

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Abstract

Sparse mixture-of-experts (MoE) routers map continuous score profiles to discrete expert sets, yet endpoint comparisons obscure when routing changes and which organizational scale changes first. We follow one Qwen3-30B-A3B model through a controlled 1,000-step reasoning-SFT run, saving 15 checkpoints and teacher-forcing the same 64 traces (36,853 tokens) at every state. The central finding is a separation of temporal profiles. By step 50, selected-set, rank-window, and shared-ID rank drift have already reached 76.6%, 85.0%, and 79.6% of their step-1,000 displacement, whereas effective expert count and Gini have reached only 28.2% and 26.7% of their endpoint changes. Local turnover then persists through $900 \rightarrow 1000$ despite slower net displacement. Thus aggregate allocation and token-level competition are not interchangeable views of one scalar routing state. A held-out 128-prompt audit also shows that capped exact-match accuracy does not improve monotonically with routing drift, while output length and censoring rise. Cross-sectional replay across Qwen, Kanana, Nemotron, and Gemma4 supports the measurement decomposition, and a separate margin-erosion intervention validates local transition vulnerability. These supporting studies bound the main claim: one controlled trajectory reveals early local reorganization followed by slower aggregate redistribution, not a universal training law or a causal account of capability.

1 Introduction

Sparse MoE routers convert a continuous score profile into a discrete top-k expert set at every token (Shazeer et al., 2017; Lepikhin et al., 2021; Fedus et al., 2022). Existing endpoint comparisons can show that two models route differently, but not *when* that difference formed or whether aggregate expert use and local expert competition followed the same training trajectory. We address this gap with a controlled full-parameter reasoning-SFT run. At Base and 15 checkpoints, the same token IDs are replayed and routing is measured at three resolutions: aggregate allocation, local selected/rank-window organization, and cutoff margin.

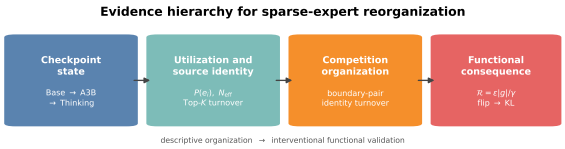


Figure 1: Evidence hierarchy. The controlled same-content trajectory asks when and at what resolution routing changes. Cross-sectional checkpoints and model families test scope. Local interventions separately test the vulnerability and consequence of the resulting routing states.

The paper has one primary claim: *routing reorganization is multi-resolution and its resolutions follow different empirical temporal profiles*. Most net local identity and rank change is already present at the earliest observed checkpoint, step 50, whereas aggregate allocation continues to redistribute through step 1,000. Non-zero adjacent-checkpoint turnover further shows that a nearly flat Base-relative curve is not a stationary router. Figure 1 separates this trajectory claim from two supporting questions: whether the decomposition appears outside the controlled run, and whether a specified local competition is vulnerable to intervention.

Our contributions are: (i) a same-content, 15-checkpoint routing trajectory; (ii) direct evidence that local and aggregate routing observables evolve on different empirical schedules; and (iii) supporting scope and intervention tests that do not conflate reorganization with capability or vulnerability.

1.1 Prior work on MoE reorganization

Routing, balance, and stability in sparse MoEs. Sparse gating, scalable routing, and balance objec-

tives establish load control and assignment stability as central MoE concerns (Shazeer et al., 2017; Lepikhin et al., 2021; Fedus et al., 2022; Lewis et al., 2021; Roller et al., 2021; Zoph et al., 2022; Dai et al., 2022; Wang et al., 2024). Our question is narrower: how aggregate load, selected identity, rank-window membership, shared-ID rank, and margin evolve along one fixed-content SFT trajectory.

Expert specialization and routing interpretation. Expert specialization is not guaranteed or reducible to one semantic label (Krishnamurthy et al., 2023; Avinash, 2026; Wang et al., 2026; Lasy et al., 2026). We study a complementary axis: stability of aligned local competitions across training states. Same-content replay separates this from generated-text change, and cross-fitted permutations test whether global expert renaming suffices.

Fine-tuning and functional interventions. Fine-tuning analyses, router-stability work, and counterfactual expert interventions motivate studying functional consequences (Tay et al., 2022; Zhang et al., 2026; Yoon et al., 2026; Lai et al., 2025). Our intervention is secondary to the trajectory: it scales one selected post-gating contribution and asks whether a locally normalized margin derivative predicts a specified downstream transition. Fisher geometry motivates the normalization (Amari, 1998); it is not a global importance score.

2 Method: observing and controlling MoE routing

We distinguish three levels. *Aggregate allocation* summarizes how often experts are used. *Local organization* describes the selected identities and near-boundary score order at one token and layer. *Cross-state drift* compares aligned local organizations.

Let $c_{\ell i}$ be the number of selected routes assigned to expert i at layer ℓ, and let $p_{\ell i} = c_{\ell i} / \sum_j c_{\ell j}$. We report the effective expert count and Gini concentration

$$H_\ell = -\sum_i p_{\ell i} \log p_{\ell i}, \qquad N_{\mathrm{eff},\ell} = \exp(H_\ell), \tag{1}$$

$$G_\ell = \frac{2\sum_{i=1}^{E} i p_{\ell(i)} - E - 1}{E}, \tag{2}$$

where $p_{\ell(i)}$ is sorted in ascending order. These quantities measure usage concentration, not aligned expert identity.

For router scores $r_{\ell,t,i}$, let $A_{\ell,t}$ be the selected top-k expert set and let $B_{\ell,t}$ be the fixed five-rank window spanning ranks 6–10, used as a near-cutoff diagnostic throughout the reported analyses. The local safety margin is

$$\gamma_{\ell,t} = r_{\ell,t,(k)} - r_{\ell,t,(k+1)}. \tag{3}$$

For cross-state comparisons at aligned sites from checkpoints or rendered contexts a and b, we report

$$J_A(a,b) = \frac{|A^a \cap A^b|}{|A^a \cup A^b|}, \qquad D^A(a,b) = 1 - J_A(a,b), \tag{4}$$

$$J_B(a,b) = \frac{|B^a \cap B^b|}{|B^a \cup B^b|}, \qquad D^B(a,b) = 1 - J_B(a,b), \tag{5}$$

$$D^{\mathrm{rank}}(a,b) = 1 - \rho_s(r_I^a, r_I^b), \qquad I = I_{16}^a \cap I_{16}^b. \tag{6}$$

We additionally report the geometric margin ratio

$$M(a,b) = \exp\left[\operatorname*{median}_{\ell} \mathbb{E}_{x,t} \log \frac{\gamma_{\ell,t}^b}{\gamma_{\ell,t}^a}\right]. \tag{7}$$

These observables are distinct observables: the scalar margin measures only the rank-k/rank-$(k+1)$ gap, while rank and set metrics measure broader score ordering and identity. Cross-family score values are never treated as calibrated on a common numerical scale. For top-8 Qwen and Gemma, this window spans ranks on both sides of the rank-8/rank-9 boundary; for top-6 Kanana and Nemotron, it contains rank 6 and four ranks outside the selected set. It is therefore not a symmetric boundary neighborhood in every architecture, and cross-family results are interpreted as complementary rank-position diagnostics rather than directly equivalent boundary-neighborhood measurements. We use the tuple

$$\mathcal{C}(x,\ell,t) = \big(A(x,\ell,t), B(x,\ell,t), r_B(x,\ell,t), \gamma(x,\ell,t)\big) \tag{8}$$

as shorthand for local competition organization. The paper compares $\mathcal{C}^{\mathrm{Base}}(x,\ell,t)$ with post-trained states at aligned contexts and sites; it does not assume that one global expert map $E_i^{\mathrm{Base}} \mapsto E_j^{\mathrm{Post}}$ is sufficient. Here r_B denotes scores restricted to the rank window; full retained score profiles are used only as supporting representations.

For a training step s, Base-relative drift uses $(a,b) = (0,s)$. Adjacent drift uses the previous saved checkpoint (s^-, s). Because checkpoint

spacing changes after step 500, we compare normalized adjacent drift

$$\widetilde{V}_s^{(m)} = \frac{D^{(m)}(s^-, s)}{s - s^-}. \tag{9}$$

The sum of adjacent dissimilarities is an operational cumulative statistic; in particular, we do not assume that $1 - \rho_s$ defines a metric-space path.

2.1 Supporting functional assay

For a selected expert contribution $u_i = g_i(h) f_i(h)$, we apply a temporary post-gating scale $u_i \mapsto (1 + \delta_i) u_i$ and allow all downstream routers to execute naturally. The source site (ℓ_s, t_s) is the selected contribution at source layer ℓ_s and the token containing the question's last non-punctuation content character. A target routing site (ℓ_d, t_d) may be any monitored later-layer position from t_s through the final prompt token; the next-token distribution is evaluated only at that final prompt position. For the target site's rank-k/rank-$(k+1)$ margin γ_{ℓ_d, t_d}, we differentiate the baseline margin with respect to this scale and define

$$g_{i \to \ell_d, t_d} = \left. \frac{\partial \gamma_{\ell_d, t_d}}{\partial \delta_i} \right|_0, \qquad \delta_i^* = -\epsilon \operatorname{sign}(g_{i \to \ell_d, t_d}), \tag{10}$$

We choose the source-perturbation sign to decrease the positive baseline margin to first order, so $\gamma_{\ell_d, t_d}(\delta_i^*) \approx \gamma_{\ell_d, t_d}(0) - \epsilon |g_{i \to \ell_d, t_d}|$. Thus $\mathcal{R}$ measures predicted margin erosion relative to the available gap, and $\mathcal{R} \geq 1$ predicts first-order boundary crossing. The intervention changes neither parameters nor the source router's selected IDs. $\mathcal{R}$ is a baseline-only vulnerability assay for a specified downstream boundary, not a definition of reorganization or expert importance. We prospectively compare the highest-risk source with matched-low and random active sources at the same destination; separately, on records that actually flip, we restore baseline expert IDs and measure the recovered next-token KL. Full algebra, numerical audits, selection rules, and negative response-geometry results are in Appendix C.

2.2 Experimental protocol

Models and execution. The controlled trajectory starts from Qwen3-30B-A3B-Base and is the primary experiment. Cross-sectional triangulation adds Qwen3-30B-A3B and Qwen3-30B-A3B-Thinking-2507. Qwen has 48 MoE layers, 128 experts, and top-8 routing. Cross-architecture score audits add Kanana-2, Nemotron-3-Nano, and Gemma 4 Base/post-trained conditions using each model's native ranking score and top-k. Expert IDs are compared only within families. The functional assay uses one middle source layer and all later MoE routers; family-specific layer, shared-expert, rendering, and numerical details appear in the appendix. Intervention models run in FP32 with FP64 KL evaluation.

Controlled trajectory. We full-parameter fine-tune Qwen3-30B-A3B-Base on 32,000 verified reasoning traces frozen from `open-r1/OpenR1-Math-220k`. Deduplication, trace verification, the 8,192-token no-truncation filter, and all hashes are reported in Appendix N. Training uses BF16 Adam, a cosine 10^{-5}–10^{-6} schedule with 5% warmup, global batch 32, fixed data order and seed, and router auxiliary coefficient .001. We save every 50 steps through 500 and every 100 steps thereafter. Full optimizer, scheduler, and RNG state are restored at each resume.

At Base and all 15 checkpoints, we teacher-force the same 64 stored Thinking-generated traces (36,853 prompt-plus-continuation tokens) and retain top-16 scores and IDs. Trajectory statistics use generated-continuation tokens. This fixed-content design removes generation differences from the checkpoint comparison.

Behavior and supporting studies. The behavior audit uses 128 GSM8K training-split prompts held out from the SFT corpus (Cobbe et al., 2021): greedy decoding, 512 new-token maximum, and last-number exact match. All 16 states pass record-count and unique-ID checks. Separate 512-prompt atlases, Qwen Base/A3B/Thinking replay, three additional MoE families, and two prospective 128-prefix intervention splits provide supporting scope and functional evidence. These datasets and their aggregation scopes are kept separate; full split lineage, rendering details, and numerical gates are in the appendix.

Metrics and uncertainty. For the controlled trajectory, baseline-relative drift at step s is $D_s^{\text{top8}} = 1 - J(A^0, A^s)$, $D_s^{\text{win}} = 1 - J(B^0, B^s)$, and $D_s^{\text{rank}} = 1 - \rho_s(0, s)$. Adjacent drift substitutes the preceding checkpoint for Base and is normalized by interval width when needed. Selected expert counts are pooled within each layer and normalized to p_i, with $N_{\text{eff}} = \exp(-\sum_i p_i \log p_i)$ and sorted-

Table 1: Official Qwen checkpoints on the 512-prompt atlas. Aggregate use is similar while aligned selected sets and boundary pairs reorganize.

State	N_{eff}	Exact top-8 vs. Base	Boundary pair vs. Base
Base	50.0	1.000	1.000
A3B	51.9	.135	.026
Thinking	49.3	.176	.036

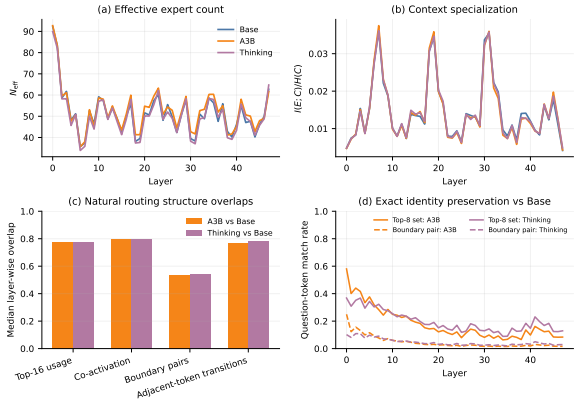


Figure 2: Official Qwen Base/post-trained observation. Aggregate expert use is similar, whereas token-level selected sets and boundary pairs turn over.

Table 2: Qwen Common-Trace Replay relative to Base. Values are layer medians.

State	Segment	Top-8 Jaccard	Shared-ID ρ_s	γ ratio
A3B	Early	.654	.753	.995
A3B	Late	.631	.727	1.059
Thinking	Early	.623	.727	1.040
Thinking	Late	.581	.687	1.043

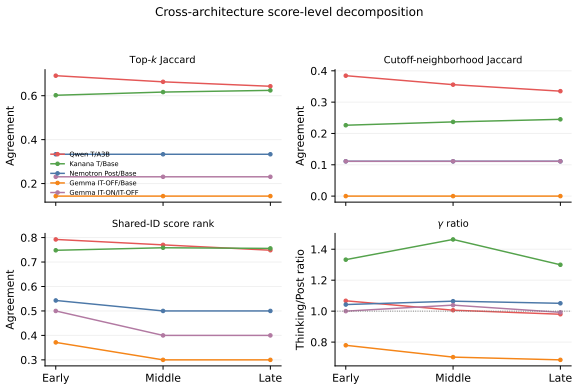


Figure 3: Official within-family score decomposition across four MoE families. Score rank, margin, rank-window membership, and selected identity are separate observables; cross-family score levels are not directly calibrated.

probability Gini $[2\sum_i ip_{(i)} - 129]/128$. Margin ratios are aggregated in log space. All headlines are medians over 48 layers; uncertainty uses 20,000 paired trace bootstraps, never treating tokens as independent. Because the observables have different scales, only their within-observable temporal profiles are compared.

3 Observation: official Base and post-trained models

We first use official checkpoints to establish what must be measured. On a 512-prompt Qwen atlas, Base, A3B, and Thinking have similar median effective expert counts (50.0, 51.9, 49.3), yet complete top-8 identity is preserved relative to Base on only 13.5% and 17.6% of token sites. Exact rank-8/rank-9 boundary-pair preservation is lower still, 2.6% and 3.6%. Aggregate use can therefore hide local reorganization.

Common-Trace Replay removes generated-text variation by teacher-forcing the same 64 Thinking-generated sequences through Qwen Base, A3B, and Thinking. Thinking's top-8 Jaccard with Base falls from .623 to .581 from early to late positions, and shared-ID rank agreement from .727 to .687, while its margin ratio remains near 1.04. Rank, selected identity, and scalar margin are therefore separable. Kanana, Nemotron, and Gemma4 show the same measurable decomposition but not a universal magnitude or direction (Figure 3).

A route-optimized global expert permutation is 88% identity and changes held-out top-8 Jaccard by only .00011. Thus neither aggregate load nor one global expert renaming recovers the local routing object. A supporting intervention additionally shows that high-risk sources enrich targeted boundary transitions and that restoring changed IDs suppresses next-token divergence (Figure 4); this is a vulnerability result, not a training-dynamics result.

4 Training: controlled Qwen trajectory

4.1 Training and fixed-content replay

Starting from Qwen3-30B-A3B-Base, we run 1,000 full-parameter BF16 SFT steps on 32,000 verified reasoning traces. We save 15 checkpoints and replay the same 64 traces, containing 36,853 prompt-plus-continuation tokens, at every state. This turns the endpoint observation above into a temporal measurement.

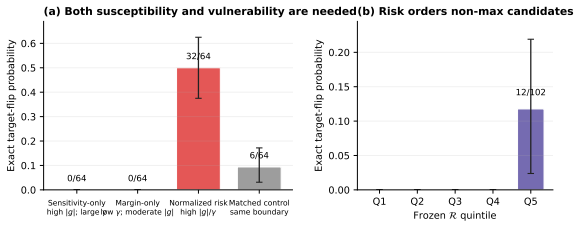


Figure 4: Supporting local control. Normalized margin erosion enriches a frozen target transition, and routing restoration suppresses divergence among records that transition.

Table 3: The main result at three organizational scales. "Step-50 share" is the fraction of the step-1,000 endpoint change already present at the earliest saved checkpoint.

Observable	Base	Step 1,000	Step-50 share
N_{eff}	66.60	89.76	28.2%
Gini	.601	.456	26.7%
Top-8 drift	.000	.453	76.6%
Rank-window drift	.000	.763	85.0%
Shared-ID rank drift	.000	.321	79.6%
Margin ratio γ_s/γ_0	1.000	.636	50.5%

4.2 Local organization changes before aggregate allocation

Under fixed optimization and identical replay content, effective expert count increases from 66.60 to 89.76 and Gini falls from .601 to .456 by step 1,000. Baseline-relative top-8, rank-window, and shared-ID rank drift reach .453, .763, and .321, while the geometric cutoff-margin ratio contracts to .636.

Step 50 already accounts for 76.6% [75.7,77.5], 85.0% [84.4,85.6], and 79.6% [78.2,81.0] of the respective endpoint local drifts, but only 28.2% [25.5,31.1] and 26.7% [24.3,29.4] of the effective-count and Gini changes. Margin contraction is intermediate at 50.5% [48.7,52.2]. Adjacent-state top-8/rank-window/rank drift is .347/.649/.256 for $0 \to 50$ and, after normalizing the wider final interval, .054/.120/.031 per 50 steps for $900 \to 1000$. Observed drift therefore decelerates but persists; it does not establish convergence; individual intervals exhibit small fluctuations. The summed adjacent-checkpoint dissimilarities are 2.458/5.327/1.525, substantially larger than the corresponding endpoint dissimilarities .453/.763/.321, showing repeated local turnover.

Because step 50 is the earliest saved checkpoint, we cannot localize its change within the first 50 updates. Layer-by-step maps show broad depth

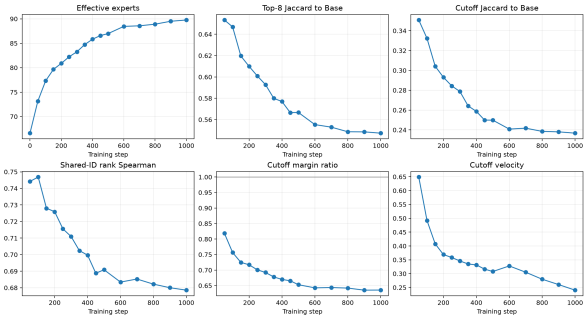


Figure 5: Primary controlled trajectory under identical replay content. Local organization undergoes most net endpoint drift by step 50, aggregate allocation changes more gradually, and adjacent-state turnover persists through step 1,000.

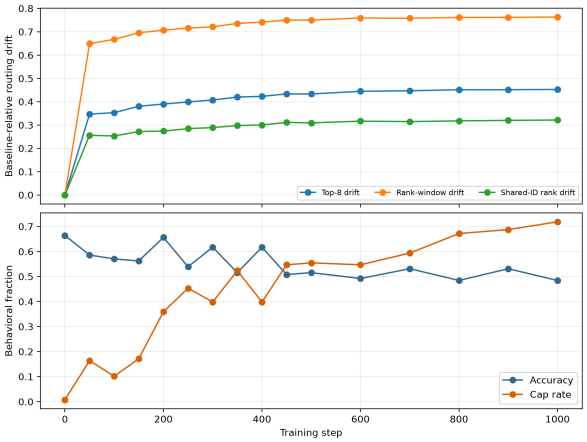


Figure 6: Routing and capped behavior along the same trajectory. Routing drift rises rapidly and then changes slowly; exact match fluctuates while the 512-token cap rate grows. Checkpoints are descriptive states of one run, not independent samples.

involvement rather than a narrow layer sequence (Appendix N). The complete behavior audit is non-monotonic in accuracy while mean length and the 512-token cap rate increase. From Base to step 1,000, correct answers fall from 85 to 62 of 128, mean length rises from 59.1 to 426.3 tokens, and cap rate rises from .008 to .719. Because 92/128 endpoint generations are right-censored, the audit does not identify the relationship between routing reorganization and uncensored reasoning capability; it only shows that capped accuracy does not improve monotonically with routing drift. Full routing-checkpoint and behavior values appear in Appendix N.

4.3 Behavior is not the same temporal profile

The held-out capped behavior audit is summarized in Figure 6. Accuracy fluctuates while output

length and censoring rise, so the audit does not validate routing drift as a monotonic positive capability proxy.

5 Conclusion

The controlled run supports one sharply delimited conclusion: aggregate expert allocation and token-level competition follow different empirical temporal profiles. "Multi-timescale" is descriptive—we fit no dynamical constant and make no convergence claim. Same-content replay is essential: the separation is not produced by comparing different generated texts. Persistent adjacent-state turnover further means that small additional Base-relative displacement does not imply a stationary router.

Two distinctions keep this result interpretable. First, aggregate summaries do not replace the site-conditioned object $\mathcal{C}(x, \ell, t)$; neither tested global expert relabeling recovers held-out local routes. Second, organization and vulnerability answer different questions. $\mathcal{C}$ describes the competition present, whereas $\mathcal{R}$ measures how easily a specified perturbation crosses one boundary. The cross-family and intervention results support these distinctions, but do not turn the single Qwen trajectory into a universal training law.

The behavior audit supplies a boundary rather than a capability claim. Capped accuracy does not improve monotonically with routing drift, while length and right-censoring rise; 71.9% of endpoint outputs hit the cap. This establishes temporal non-equivalence under the tested decoder, not independence or causal separation between routing and reasoning ability.

Summary. One controlled reasoning-SFT trajectory reveals a two-stage empirical picture: local selected-set and rank organization reorganize early, while aggregate expert allocation continues to redistribute. Local turnover also persists after net displacement slows. Therefore, aggregate load, local competition, behavior, and perturbation vulnerability should not be collapsed into one scalar notion of "routing change." Cross-family replay and local interventions support the measurement framework, while cross-seed, cross-task, uncensored behavioral, and trajectory-level causal generalization remain open.

Limitations

The controlled evidence is one Qwen reasoning-SFT trajectory, one seed, and one task family. It supports a within-run temporal description, not a universal post-training law or causal attribution of behavioral change to routing. Step 50 is the earliest saved intermediate checkpoint, so substantial change observed there cannot be localized to particular earlier updates. Checkpoint intervals also widen after step 500; we normalize adjacent drift when comparing intervals and do not claim convergence or a fitted dynamical time constant.

The behavior audit on prompts held out from the SFT corpus is complete through step 1,000, but more than half of later generations reach the 512-token cap. It therefore measures capped-decoding behavior rather than an uncensored accuracy endpoint and does not establish independence between routing and behavior. It reuses the first prospective targeted-intervention prompt split, so the behavioral and functional analyses are not independent at the prompt-sample level, although they measure different outcomes. This split is disjoint from the earlier Stage-2 analysis set. Common-Trace Replay fixes content but conditions Base and A3B on Thinking-generated sequences they need not produce naturally. Cross-family comparisons are weaker normalized-position analyses with architecture-specific scores and tokenizers; numerical levels are not directly comparable.

The global-relabeling result applies only to the tested cross-fitted Hungarian controls. It does not exclude context-dependent correspondences or changes in experts' full functions. Functional interventions act at one source layer and evaluate downstream router events at potentially different token positions and the immediate next-token distribution at the final prompt position. Prospective sources are selected from baseline geometry rather than randomized, and routing restoration conditions on observed flips. Nemotron's multi-step derivative audit fails, so its within-model enrichment does not support a derivative-shift mechanism. The small reorganization–risk pilot does not establish absence of an interaction. Further tasks, seeds, training trajectories, layers, and uncensored behavior remain open.

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A Disclosure and reproducibility statements

Writing assistance. Generative AI tools assisted experimental code development, debugging, workflow documentation, analysis scripts, figure and table generation, and drafting and editing this manuscript. The authors reviewed AI-assisted outputs against saved configs and artifacts, executed numerical consistency checks, and retained negative results. No generative AI system is an author. The human authors take responsibility for the final content, claims, code, and artifacts.

Reproducibility. Supplementary code and artifacts specify checkpoint, pinned dataset revision, prompt and target-token protocol, split hashes, intervention definition, precision, seeds, and acceptance checks. One script regenerates every figure and result table from saved records. Appendices report derivative protocol lineage, negative gates, prospective selection hashes, routing-clamp validation, prefix-bootstrap trend intervals, and estimator uncertainty. The controlled trajectory manifest additionally records all 15 checkpoint paths

and hashes, resume lineage, fixed trace hashes, and the exact/normalized SFT-overlap audit; all route streams pass content and score-order validation. All 16 behavior states through step 1,000 pass record-count and unique-ID checks. The reserved confirmation set is excluded.

B Cross-checkpoint and cross-architecture evidence

B.1 Cross-sectional routing provides external triangulation

We record the selected expert IDs and normalized top-8 weights at every question token, every layer, and all 512 prefixes for Qwen Base, A3B, and Thinking: 35,368 prompt tokens and 13.58 million route assignments per checkpoint. Figure 7 shows three levels of organization. First, layer-wise effective expert count is nearly unchanged: its median is 50.0, 51.9, and 49.3. Second, unsupervised eight-cluster context specialization is also stable, with median normalized $I(E;C)$ of .01270, .01258, and .01255. Thus the atlas does not support the hypothesis that Thinking globally uses fewer experts or becomes more specialized on these prompt tokens.

Reorganization instead appears in identity and competition. Versus Base, the median layer-wise top-16 usage Jaccard is .778 for both A3B and Thinking, and top-1 expert identity is preserved on 64.7% and 69.4% of question tokens. The full top-8 set is preserved on only 13.5% and 17.6%, while the exact rank-8/rank-9 boundary pair is preserved on only 2.6% and 3.6%. Population edge sets are more stable but show the same ordering: median top-1000 Jaccard is .798/.794 for co-activation, .769/.779 for adjacent-token transitions, and only .532/.545 for boundary-pair competition. Post-trained states therefore preserve coarse load and much of the co-activation/transition skeleton while substantially reorganizing token-level membership and especially the marginal competition. A3B is not uniformly closer to Base than Thinking, so the atlas does not support a simple monotone training progression.

Generated-trajectory context. We greedily generate 64 Qwen Thinking continuations and use their stored token sequences for the Common-Trace experiment below. Descriptively, routing remains dynamic: median effective expert count increases from 60.8 to 68.2 from early to late observed positions, and lag-8 top-8 Jaccard (.156) remains above a matched within-trajectory shuffle (.132). Because most traces hit the 512-token cap and only Thinking is generated naturally, the full trajectory analysis is supporting rather than a checkpoint-level claim (Appendix).

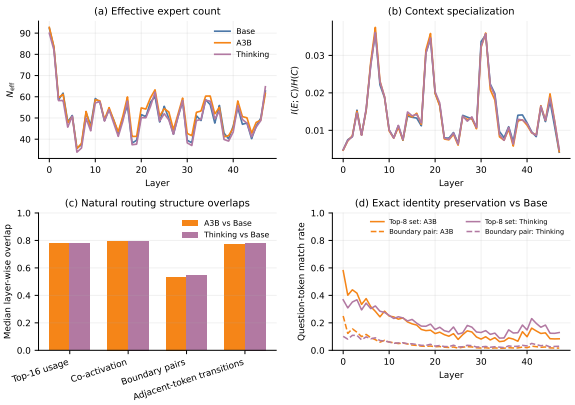


Figure 7: Natural-routing atlas over 512 GSM8K prompts. (a) Effective expert count and (b) checkpoint-independent context specialization are nearly unchanged. (c) Layer-wise population edge overlap remains highest for co-activation and adjacent-token transitions and lowest for boundary pairs. (d) Exact question-token expert identity rapidly diverges with depth, with the rank-8/rank-9 competing pair least preserved.

B.2 Same-content replay isolates checkpoint-associated organization

Natural generation confounds checkpoint identity with generated text. We remove that content mismatch by teacher-forcing the same 64 Thinking-generated token sequences through Base, A3B, and Thinking. This is a routing-response comparison under shared content, not a claim that Base or A3B would naturally produce those sequences. Substantial checkpoint differences remain. Relative to Base, median-across-layer top-8 Jaccard for A3B is .654, .643, and .631 in early, middle, and late generation; Thinking is less aligned at .623, .603, and .581. Its decline from early to late is also larger (.042 versus .023). Top-1 match remains around .61–.65 and does not show a monotone stage effect (Table 8).

The rank window shows particularly low agreement. Rank-6:10-window Jaccard changes from .334 to .322 for A3B and from .270 to .252 for Thinking. Exact rank-8/rank-9 pair agreement is only 1–2%, but we treat that as supplementary because exact pairs have high cardinality. Length-normalized curves corroborate the stage summaries: Thinking remains less aligned than

Table 4: Score-level Common-Trace decomposition with Base. Values are medians across layers for the indicated generation segment. γ ratio is the geometric mean across site-level ratios; score-rank agreement uses shared expert IDs in the top-16 lists.

Checkpoint	Segment	γ ratio	Score-profile distance	Shared-ID ρ_s	Window Jaccard
A3B	Early	0.995	.172	.753	.334
	Middle	1.056	.170	.742	.335
	Late	1.059	.173	.727	.322
Thinking	Early	1.040	.179	.727	.287
	Middle	1.047	.175	.704	.282
	Late	1.043	.177	.687	.272

A3B and reaches .566 top-8 Jaccard in the final bin, versus .616 for A3B. Therefore, holding prompt and continuation tokens fixed, checkpoint identity is associated with routing reorganization, and the Thinking state differs more from Base than ordinary A3B on these traces, especially at later observed positions.

B.3 Score-level decomposition of rank-window reorganization

Identity turnover alone does not say whether the router scores moved or merely reshuffled their local order. We therefore rerun the same-content replay while retaining the top-16 router scores (the existing ID-only replay is unchanged). For each checkpoint we report the top-8/top-9 margin γ, the centered and scale-normalized score-profile distance at ranks 6–10, and Spearman agreement of score values for expert IDs appearing in both top-16 lists. The latter is computed on shared expert IDs; Spearman on the already sorted rank slots would be tautologically one and is omitted.

Figure 8 gives the position-dependent decomposition. Thinking has geometric-mean cutoff-margin ratios of 1.040, 1.047, and 1.043 relative to Base in early, middle, and late generation, respectively. A3B has 0.995, 1.056, and 1.059, so its margin shift is less uniform. At the same time, Thinking's shared-ID score-rank Spearman falls from .727 to .704 to .687, and its rank-6:10 Jaccard falls from .287 to .282 to .272. The corresponding A3B values are .753, .742, .727 and .334, .335, .322. Thus Thinking's stronger late discrete turnover coexists with a modest margin expansion and progressive local rank reshuffling; the identity result is not explained by a single scalar margin shift. Raw score levels are interpreted only as same-architecture, same-content relative quantities because router-score parameterizations need not be globally calibrated across checkpoints.

The third edge of the Qwen checkpoint triangle

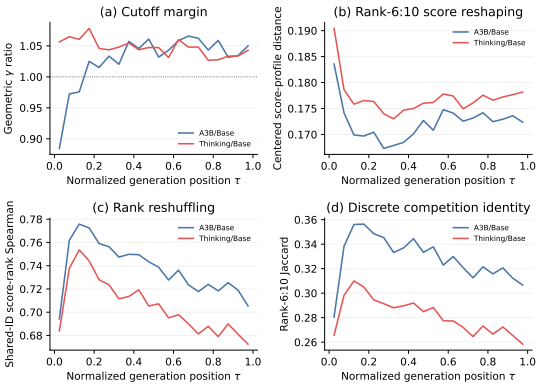


Figure 8: Score-level decomposition under Common-Trace Replay. (a) Thinking shows a modest, position-stable cutoff-margin expansion relative to Base, while A3B is less uniform. (b) Centered rank-6:10 score profiles are reshaped in both comparisons. (c) Spearman agreement of scores for expert IDs shared by the two top-16 lists declines with position, more strongly for Thinking. (d) The same positions exhibit discrete rank-window turnover. Rank-slot Spearman is excluded because both score vectors are sorted by construction.

is available without another forward pass. Direct Thinking–A3B agreement also declines from early to late: top-8 Jaccard falls from .691 to .643, rank-window Jaccard from .384 to .335, and shared-ID score-rank Spearman from .792 to .748. In contrast, the geometric cutoff-margin ratio changes only from 1.067 to .980. Thus the result is not only two separate distances from Base: A3B and Thinking themselves become less aligned over the observed continuation even though their scalar cutoff margins remain similar.

B.4 A global expert permutation does not explain turnover

Low same-ID overlap could arise if post-training merely permuted a stable set of expert roles. We test this alternative with two layer-wise Hungarian assignments, each learned on 32 prefixes and evaluated on the disjoint half. Functional- fingerprint matching improves held-out routing-profile cosine for Thinking–Base from .829 to .901, yet top-8 and rank-window Jaccard fall from .613 to .366 and .298 to .163. Thus expert-level routing correspondence is not token-level route correspondence. More directly, a second assignment optimizes binary top-8 activation-vector Jaccard on the training tokens. It is 88% fixed points, gains only about 5×10^{-4} over identity on its training objective, and changes held-out top-8 Jaccard by only .00011. The routing-optimal global permu-

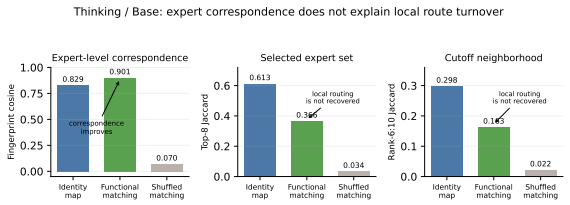


Figure 9: Three-level Qwen Thinking/Base permutation control. Hungarian assignments are learned from expert routing fingerprints on one prefix half and evaluated on the other. Functional matching raises held-out fingerprint cosine (.829 to .901), confirming expert-level correspondence, but lowers top-8 and rank-window overlap (.613 to .366 and .298 to .163). Shuffled matching shows the random-permutation expectation. Thus correspondence improves without recovering local routing. Separately, the routing-optimized map is nearly identity (88% fixed points) and changes held-out top-8 Jaccard by only .00011.

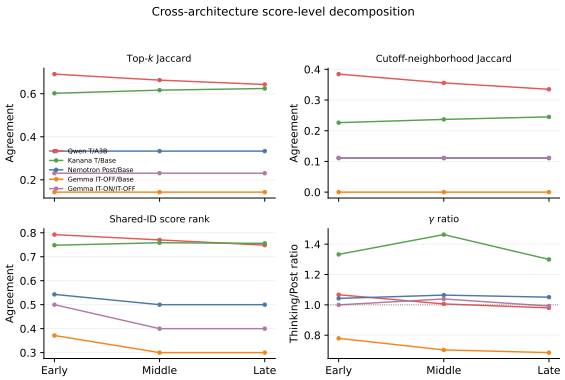


Figure 10: Cross-architecture score-level decomposition on fixed prompt strings. Qwen and Kanana use the generated/common or plain replay described above; Nemotron and Gemma use normalized-position matching with checkpoint-specific tokenizers. Across families, score-rank reshaping and rank-window identity are measurable separately, while margin magnitude and positional profiles are heterogeneous.

tation is therefore nearly identity and provides no substantive recovery. Figure 9 shows the primary Thinking–Base control; complete pairs, intervals, and objectives are in the appendix. These routing-derived fingerprints are not evidence of semantic or full input–output equivalence.

B.5 Cross-architecture score-level decomposition

Kanana. We next repeat only the four most diagnostic measurements in an independent MoE architecture. We teacher-force the same 64 plain GSM8K prompt strings through Kanana Base and Thinking, using their native tokenizer and actual choice score $r_i = \sigma(q_i) + b_i$. Kanana has top-6 routing, so γ is the rank-6/rank-7 gap; ranks 6–10 form the fixed five-rank diagnostic window, with only rank 6 on the selected side. This replay contains 4,409 prompt tokens across 47 MoE layers and performs no intervention.

Table 6 shows that Kanana also has low rank-window identity agreement (.226–.245) despite higher top-6 set overlap (.602–.624), together with shared-ID score-rank agreement around .75. Its margin ratio is larger and nonmonotone (1.333, 1.463, and 1.300), and its identity/rank metrics do not exhibit Qwen's late-position decline. The independent family therefore reproduces the continuous-score/discrete-identity decomposition but not a universal positional direction. We interpret this as architecture- and checkpoint-dependent heterogeneity, not a failed replication.

Nemotron and Gemma4. To test whether the decomposition was specific to the two preceding families, we repeat the same fixed-prompt normalized-position analysis for the existing Nemotron Base–Post checkpoints and for Gemma4 Base, IT-OFF, and IT-ON. Nemotron uses its corrected choice score $\sigma(q) + b$ and top-6 routing; Gemma uses the FP32 pre-top-8 softmax probability. Each comparison contains 64 identical prompt strings, but checkpoint-specific tokenizers, so these are normalized position comparisons rather than token-identity common traces.

The results in Table 5 and Figure 10 reinforce the heterogeneous interpretation. Nemotron Post/Base has top-6 Jaccard .384, .348, and .326 and rank-window Jaccard .146, .132, and .123 from early to late, while its margin ratio remains close to one (1.043, 1.064, 1.051). Gemma IT-OFF/Base has very low local identity agreement (top-8 Jaccard .143–.148 and rank-window Jaccard about .055) and a margin ratio that falls from .779 to .685. Holding Gemma IT weights fixed, IT-ON/IT-OFF has margin ratios 1.000, 1.039, and .992; its score-rank and rank-window identity profiles shift, but neither follows the Qwen late-decline pattern. Thus the robust cross-family statement is that continuous score profiles and discrete rank-window identities can be analyzed as separate quantities; their amplitude and positional direction depend on architecture, checkpoint, and rendered context.

The positional contrasts are not inferred from point estimates alone. A prefix-bootstrap over the

Table 5: Nemotron and Gemma4 score-level replication. Entries are medians over router layers for early/middle/late normalized positions. Top-k is 6 for Nemotron and 8 for Gemma; rank-6:10 is the fixed five-ID diagnostic window. It is not symmetric around the top-6 boundary.

Pair	Segment	γ ratio	Shared-ID ρ_s	Window Jaccard	Top-k Jaccard
Nemotron Post/Base	Early	1.043	.543	.146	.384
	Middle	1.064	.500	.132	.348
	Late	1.051	.500	.123	.326
Gemma IT-OFF/Base	Early	.779	.371	.056	.143
	Middle	.703	.300	.056	.143
	Late	.685	.300	.054	.143
Gemma IT-ON/IT-OFF	Early	1.000	.500	.111	.231
	Middle	1.039	.400	.111	.231
	Late	.992	.400	.111	.231

Table 6: Direct Qwen Thinking–A3B comparison and minimal Kanana replication. Values are medians across layers. γ ratios are Thinking/A3B for Qwen and Thinking/Base for Kanana; top-k is 8 and 6, respectively.

Pair	Segment	γ ratio	Shared-ID ρ_s	Rank-window Jaccard	Top-k Jaccard
Qwen Think./A3B	Early	1.067	.792	.384	.691
	Middle	1.006	.770	.356	.663
	Late	.980	.748	.335	.643
Kanana Think./Base	Early	1.333	.748	.226	.602
	Middle	1.463	.758	.237	.617
	Late	1.300	.756	.245	.624

64 replay prefixes gives the early-minus-late intervals in Table 7. For Qwen, all three identity/rank contrasts exclude zero. Kanana has a small early-to-late increase in overlap for the two identity measures, while its score-rank agreement and margin ratio are nonmonotone; the data therefore do not support a common monotone positional trajectory.

Supporting global-summary tests. Earlier experiments tested whether a single full response matrix or stationary source-to-router graph could replace the local description. In a 16-prefix pilot, full and diagonal response predictors were similar ($R^2 = .837$ and .841); over 128 prefixes, pooled full-matrix R^2 fell to $-.088$. A preregistered static graph also failed edge-rank, top-edge, and out-strength stability gates. These negative results motivate local measurement but are not a second headline; full geometry, estimator, and graph diagnostics appear in the appendix.

C Supporting response geometry

For prefix x, let $p_0(y \mid x)$ and $p_\delta(y \mid x)$ be baseline and intervened next-token distributions. We define

$$s_i(x,y) = \left.\frac{\partial \log p_\delta(y \mid x)}{\partial \delta_i}\right|_{\delta=0}, \qquad F = \mathbb{E}_{x,y\sim p_0(\cdot|x)}[s(x,y)s(x,y)^\top] \tag{11}$$

and accumulate $\widehat{F} = N^{-1}\sum_n s_n s_n^\top$ in FP64. Within a fixed-routing region, $\mathbb{E}_x D_{\mathrm{KL}}(p_0 \| p_\delta) = \frac{1}{2}\delta^\top F \delta + o(\|\delta\|_2^2)$. We compare the full predictor with its diagonal counterpart. The small pilot supports local quadratic prediction, but the full matrix does not beat the diagonal and pooled absolute magnitude fails the expanded-prefix robustness check. We therefore treat this geometry as supporting history rather than the paper's main object.

Table 7: Prefix-bootstrap early-minus-late contrasts over 64 replay prefixes. Values are differences in the layer-median stage summaries; intervals contain 95% bootstrap replicates (20,000 draws). Positive values indicate larger early agreement.

Pair	Metric	Early − Late	95% CI
Qwen Think./A3B	Top-k Jaccard	.047	[.038,.056]
	Rank-window Jaccard	.048	[.039,.057]
	Shared-ID ρ_s	.044	[.036,.051]
Kanana Think./Base	Top-k Jaccard	$-.023$	$[-.034, -.014]$
	Rank-window Jaccard	$-.016$	$[-.027, -.007]$
	Shared-ID ρ_s	$-.004$	$[-.012, .005]$

Table 8: Common-Trace Replay agreement with Qwen Base. Values are medians across 48 layers under identical Thinking-generated token sequences.

Checkpoint	Segment	Top-8 Jaccard	Rank-window Jaccard	Top-1 match
A3B	Early	.654	.334	.640
	Middle	.643	.335	.650
	Late	.631	.322	.637
	Answer*	.618	.307	.642
Thinking	Early	.623	.270	.613
	Middle	.603	.263	.628
	Late	.581	.252	.608
	Answer*	.576	.246	.628

D Natural identity turnover

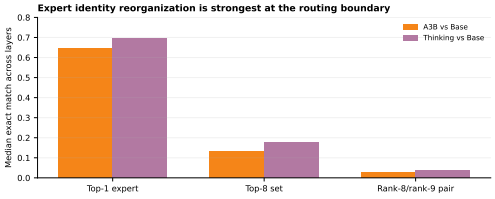


Figure 11: Median exact identity preservation across layers. The top-1 expert remains relatively stable, the full top-8 set usually changes, and the rank-8/rank-9 competing pair changes on more than 96% of natural question-token events.

E Generated-trajectory descriptive figure

F Implementation validation

Layer 47 was used for the formal finite-difference gate to minimize accumulated downstream round-off; the same adapter and algebra are used at layer 23. An initial BF16 score path was rejected because its numerical discrepancy was too large; all

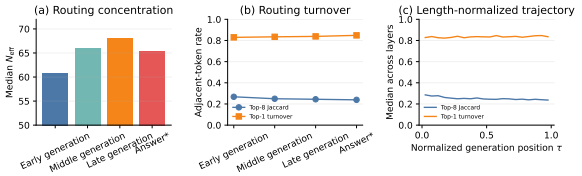


Figure 12: Routing along 64 Qwen Thinking generated trajectories. Effective expert count increases and adjacent-token top-8 overlap decreases from early to late observed generation. Most generations hit the 512-token cap; answer estimates use only delimiter-positive traces.

Table 9: Numerical and algebraic validation checks; all listed checks pass.

Check	Setting	Result
Zero intervention	layer 23, FP32	max logit difference 0
Inactive experts	layer 47, 20 cases	autograd and FD max 0
Finite difference	layer 47, 80 active	median error 0.735%
Finite difference	layer 47, 80 active	P90 error 4.942%
Finite difference	layer 47, 80 active	sign agreement 100%
Repeat score	fixed seed/prefix	max difference 0
F symmetry	$N = 512$	relative error 0
Streaming vs. offline	$N = 512$	1.57×10^{-16}
Positive semidefinite	$N = 512$	$\lambda_{\min} = -2.24 \times 10^{-20}$

Table 10: Local prediction stratified by intervened expert count.

$\lvert S\rvert$	Predictor	n	R^2	Pearson r	Spearman ρ_s
1	Full	10	−0.308	0.149	0.423
1	Diagonal	10	−0.308	0.149	0.423
2	Full	10	−1.691	−0.008	0.067
2	Diagonal	10	−1.682	−0.019	0.067
4	Full	10	0.646	0.820	0.503
4	Diagonal	10	0.682	0.836	0.503
8	Full	10	0.727	0.861	0.903
8	Diagonal	10	0.724	0.861	0.879

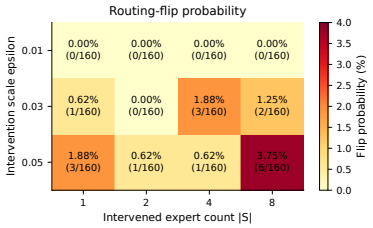


Figure 14: Flip probability by perturbation scale and subset size. Each cell contains 160 records. No record flips at $\epsilon = 0.01$. Percentages are descriptive because records share prefixes and interventions.

reported response and intervention results use strict FP32.

G Detailed local prediction results

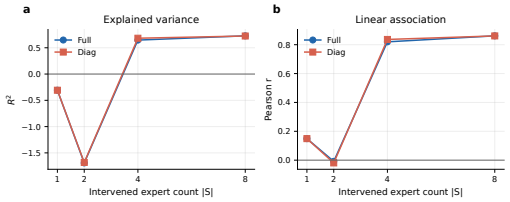


Figure 13: Descriptive local metrics by intervention subset size. Each stratum contains 10 interventions; the pooled result is primary.

Negative R^2 for small subsets reflects tiny observed KL and only 10 cases per stratum. Full and diagonal are identical for $|S| = 1$ by construction.

H Routing analysis details

I Sampling stability and estimator redesign

J Artifact scope

All figures are generated from saved JSONL, JSON, CSV, and NumPy artifacts. The pilot local prediction unit is an intervention mean over 16 prefixes; the E4 expanded unit is an intervention mean over 128 prefixes; the routing unit is one intervention–prefix record. These units remain separate to avoid inflating sample size. E4 artifacts are stored under `artifacts/followup/e4_expanded_local_v2/` and include the 20,000- replicate bootstrap results. The generation manifest records output hashes and that the redesign confirmation set was not accessed. The Qwen–A3B and Kanana score-replay trend intervals are generated from `artifacts/graph/reorganization_trend_bootstr` using 20,000 prefix resamples; no additional model forward is required.

K Derivative protocol lineage

The original stock sequence-wide G0 full-vector finite-difference gate failed. Component audits localized the discrepancy to the stock in-place expert aggregation: `torch.func.jvp` and low-level forward AD agreed with each other but not with reverse mode or stable finite differences. On the documented out-of-place `index_add` amendment, forward AD, low-level dual tensors, `autograd.functional.jvp`, reverse mode, and finite differences agree at the audited router event. Full-vector forward/reverse relative error is 1.67×10^{-6} with cosine effectively one; stable $\epsilon = 10^{-3}$ finite difference has relative error 0.0129

Table 11: Record-level routing-transition summary.

Quantity	Estimate	Uncertainty/reference
Records	1,920	16 prefixes × 120 interventions
Records with flip	17	cluster rate CI $[0.0021, 0.0203]$
Affected interventions	14	of 120
ρ ROC–AUC	0.9977	Mann–Whitney $p = 4.24 \times 10^{-16}$
Median ρ, flip	33.415	no-flip P90 $= 0.892$
Mean KL, flip	6.822×10^{-4}	no-flip $= 2.765 \times 10^{-9}$
Mean log-KL contrast	flip minus no-flip	cluster CI $[6.77, 9.62]$

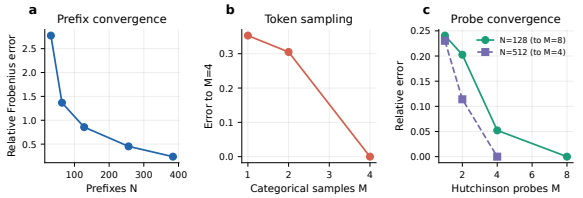


Figure 15: Sampling diagnostics: nested-prefix estimates, categorical token sampling, and Hutchinson probe convergence. Probe convergence does not imply convergence over independent prefixes.

Table 12: Prefix-split stability; lower difference and higher overlap are better.

Estimator	N	Half difference	Top-8 overlap	Decision
Categorical $M = 4$	512	0.762	0.613	unstable
Categorical RB $M = 4$	128	0.953	0.693	unstable
Hutchinson $M = 8$	128	0.892	0.564	unstable
Hutchinson $M = 4$	512	0.566	0.631	narrow gate fail

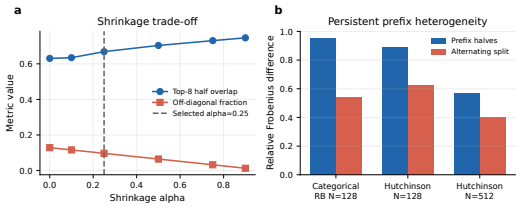


Figure 16: Exploratory redesign. Shrinkage trades subspace stability against off-diagonal magnitude, while prefix heterogeneity persists across estimators. Outcome tuning did not select a superior estimator, so confirmation stayed sealed.

and cosine 0.99992.

A separate 8-prefix primal audit verifies that the amendment changes only expert aggregation, not parameters, routing policy, intervention definition, targets, or thresholds. Small FP32 differences are reported as accumulation-order effects rather than requiring bit identity. The stock failure remains recorded; it is not retroactively relabeled. The amended scalar router-margin derivative subsequently passes its primary AD oracle gate. Position-dependent full-vector FD remains an explicit numerical diagnostic, not a replacement gate.

L Functional assay details

Prospective transition rates. Across the first two independently frozen 128-prefix splits, high-risk, matched-low, and random sources change the exact targeted event at rates .4609/.1016/.1172 and .3594/.0938/.1250. The paired high-minus-low effects are .3594 [.2812,.4453] and .2656 [.1875,.3438]; all 46 and 34 discordant pairs are high-only. These are baseline-selected predictive contrasts rather than randomized treatment effects.

Component and non-max controls. On a disjoint 64-prefix split, sensitivity-only and margin-only selectors each produce 0/64 transitions, while normalized $|g|/\gamma$ produces 32/64 versus 6/64 for a source-matched same-boundary control. On another 64 prefixes, all 12 transitions among 512 non-max candidates fall in the highest frozen-risk quintile (Table 18).

Cross-family summary. High-minus-low effects are .359/.336/.219 for Qwen Base/A3B/Thinking, .172/.148 for Nemotron Base/Post, .125/.250/.203 for Gemma Base/IT-OFF/IT-ON, and .414/.289 for Kanana Base/Thinking. Only the Qwen Thinking–Base effect contrast is resolved; the other checkpoint/mode contrasts are not. These results support transfer of within-condition enrichment, not a universal robustness shift.

Reorganization–risk pilot. The balanced 54-site $O^B \times \mathcal{R}$ pilot observes only two target transitions, both at high risk but at opposite reorganization levels. It does not resolve an interaction between boundary reorganization and vulnerability.

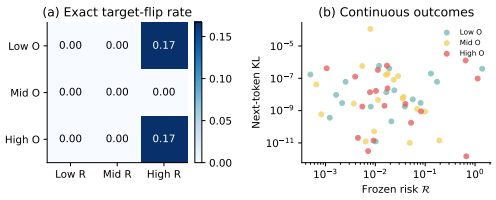


Figure 17: Balanced selected-site mechanism pilot. Exact target-flip rates and next-token KL after crossing rank-window reorganization with frozen Base risk. Only two target flips occur, both at high risk but at opposite reorganization levels; the experiment does not establish an $O^B \times \mathcal{R}$ interaction.

The retrospective routing-event audit logs 2,206,080 downstream events for 120 interventions and 128 prefixes, forming 15,360 record aggregates. It finds 551 flip events in 225 records and 68 interventions. No flip satisfies $2\|\Delta q\|_\infty < \gamma$. Among

Table 13: Exploratory no-flip shrinkage tuning smoke ($n = 11$).

Predictor	R^2	MSE	MAE
Shrinkage ($\alpha = 0.25$)	0.5042	4.1769×10^{-19}	4.5973×10^{-10}
Raw response	0.5057	4.1644×10^{-19}	4.5792×10^{-10}
Diagonal	0.4991	4.2201×10^{-19}	4.6515×10^{-10}

affected records, 145/225 flips occur at the record's minimum-margin event. Flip-event median margin is 0.003109 versus 0.091072 for nonflips; median router displacement is 0.03374. Flips concentrate in layers 45–46, followed by 42, 41, and 44. Event maxima reproduce legacy record-level ρ exactly.

Table 14: Routing restoration over all 225 observed transition records.

Clamp	Median suppression	P10	Fraction > 0.5	Fraction > 0.9
First changed event	0.99677	0.85672	0.9600	0.8756
All changed IDs	0.99677	0.86761	0.9689	0.8844
All IDs and weights	0.99661	–	–	–

First- and all-clamp KL are exactly equal in 216 records. The nine exceptions are retained. The minimum all-clamp suppression is -0.0106; one slightly negative case prevents an absolute restoration claim.

M Prospective selection and cross-architecture details

Table 15: Held-out functional-fingerprint permutation control. Entries show raw Jaccard, Jaccard after cross-fitted Hungarian relabeling, and matched-minus-raw change with 95% prefix-bootstrap CI (20,000 draws).

Pair	Set	Raw	Matched	Δ (95% CI)
A3B/Base	Top-8	.655	.437	$-.218$ [$-.222$, $-.215$]
	Rank-window	.348	.213	$-.135$ [$-.138$, $-.132$]
Thinking/Base	Top-8	.613	.366	$-.247$ [$-.251$, $-.243$]
	Rank-window	.298	.163	$-.135$ [$-.139$, $-.132$]
Thinking/A3B	Top-8	.672	.442	$-.230$ [$-.234$, $-.225$]
	Rank-window	.366	.218	$-.148$ [$-.152$, $-.145$]

Generated trajectory and Common-Trace Replay audit. The Thinking trajectory set contains 64 unique prompts, 32,344 generated tokens, and 36,853 prompt-plus-continuation tokens. Only 6/64 traces finish before the 512-token cap and 23/64 contain the closing thought delimiter. The replay grids for Base, A3B, and Thinking each contain the same 64 complete sequences and pass record-count, uniqueness, shape, normalized-weight-sum, and content-hash checks. Their record SHA256 values begin `43f83e250209bced`, `ddfcd7ea2520f97e`, and `d2a7c4b9eccc9558`. No sealed confirmation record is read. Exact rank-8/rank-9 agreement with Base is .022/.022/.022 across A3B's early/middle/late segments and .016/.015/.014 for Thinking. We retain these values for completeness but prefer rank-window Jaccard because exact pair identity is a high-cardinality statistic.

N Controlled training-trajectory details

The training trajectory uses Qwen3-30B-A3B-Base and full-parameter BF16 reasoning SFT. The frozen corpus is a seeded reservoir sample of 32,000 unique problems from the default training split of `open-r1/OpenR1-Math-220k`. We deduplicate by UUID, or by normalized problem hash when UUID is absent. Per row, we select the first generation for which both `is_reasoning_complete` and `correctness_math_verify` are true; require a nonempty problem, a closing `</think>`, and a nonempty answer span; and serialize a user problem plus the complete verified assistant trace with the Qwen chat template. Sequences longer than 8,192 tokens are filtered without truncation. Selection uses seed 20261003; the resulting JSONL order is frozen, and the reasoning arm computes loss over the complete assistant trace.

We use Adam with $\beta_1 = .9$, $\beta_2 = .95$, $\epsilon = 10^{-8}$, constant weight decay .1, micro batch 1, and global batch 32. We fix training and data-order seeds to 20261004, disable dataloader shuffling, use a cosine schedule from 10^{-5} to 10^{-6} over 1,000 planned steps with 5% warmup, and set the router auxiliary coefficient to .001. Checkpoints 50–200 belong to the initial process; checkpoints 250–500 continue from checkpoint 200, checkpoints 600–800 continue from checkpoint 500, and checkpoints 900–1000 continue from checkpoint 800. Each resume restores model, optimizer, scheduler, and RNG state. Each checkpoint contains eight distributed shards and its recorded iteration matches its directory name.

All 16 states, including Base, replay the same 64 traces and 36,853 prompt-plus-continuation token IDs. Each route stream passes record count, uniqueness, shape, descending top-16 score, normalized selected-weight sum, and content-hash checks. Neither these 64 problems nor the separate 512-prompt atlas has exact or normalized overlap with the SFT corpus. Figure 18 gives the two layer-by-step views.

Both show broad depth involvement; they do not support a simple narrow-band or sequential layer hierarchy.

Table 16: Controlled trajectory. Drift is one minus agreement with Base; values are medians across 48 layers.

Step	N_{eff}	Gini	Top-8	Rank-6:10 window	Rank	γ_s/γ_0
0	66.60	.601	.000	.000	.000	1.000
50	73.15	.563	.347	.649	.256	.819
100	77.31	.534	.353	.668	.253	.757
150	79.66	.515	.380	.696	.272	.725
200	80.88	.510	.390	.707	.274	.717
250	82.23	.501	.399	.716	.284	.701
300	83.26	.495	.407	.721	.289	.692
350	84.73	.487	.420	.736	.298	.678
400	85.83	.481	.423	.741	.300	.670
450	86.57	.476	.433	.750	.311	.666
500	86.98	.473	.433	.750	.309	.653
600	88.45	.464	.445	.759	.317	.643
700	88.57	.463	.447	.758	.315	.644
800	88.91	.461	.451	.761	.318	.642
900	89.50	.457	.451	.762	.320	.635
1000	89.76	.456	.453	.763	.321	.636

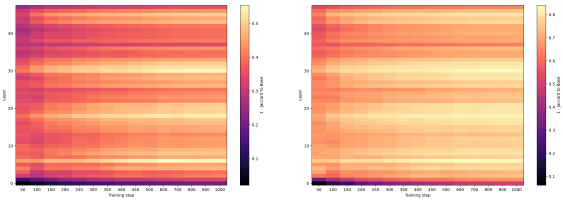


Figure 18: Layer-by-training-step baseline-relative drift. Left: selected top-8 set. Right: rank-6:10 window. The common color interpretation is $1-$Jaccard, but each panel has its own empirical scale.

The adjacent-checkpoint drifts over 50-step intervals through step 500 for top-8/rank-window/rank are: .347/.649/.256 ($0 \rightarrow 50$), .232/.492/.155 ($50 \rightarrow 100$), .187/.408/.114 ($100 \rightarrow 150$), .162/.369/.095 ($150 \rightarrow 200$), .162/.359/.097 ($200 \rightarrow 250$), .156/.346/.093 ($250 \rightarrow 300$), .146/.335/.087 ($300 \rightarrow 350$), .152/.332/.092 ($350 \rightarrow 400$), .140/.316/.082 ($400 \rightarrow 450$), and .139/.308/.080 ($450 \rightarrow 500$). For reference, at step 500 the 95% paired-bootstrap intervals for top-8 Jaccard, rank-window Jaccard, and shared-ID rank Spearman are [.561,.572], [.245,.254], and [.684,.697]; the interval for log margin ratio is [-.437,-.409]. The later 100-step adjacent-checkpoint drifts are .146/.328/.085, .136/.305/.081, .127/.280/.076, .118/.261/.071, and .108/.240/.061 for $500 \rightarrow 600$ through $900 \rightarrow 1000$. Because interval widths differ, this sequence is descriptive rather than a per-update convergence rate. The final triplet is .054/.120/.031 per 50 training steps, so observed drift decelerates but persists and does not establish convergence. At step 1,000 the 95% paired-bootstrap intervals for top-8 Jaccard, rank-window Jaccard, and shared-ID rank Spearman are [.541,.554], [.231,.240], and [.673,.685]; the interval for log margin ratio is [-.463,-.437]. Tokens contribute within a trace but are not bootstrap units.

Step 50 accounts for 76.6% [75.7,77.5], 85.0% [84.4,85.6], and 79.6% [78.2,81.0] of the final top-8, rank-window, and rank drift, compared with 50.5% [48.7,52.2] of margin contraction, 28.2% [25.5,31.1] of effective-count change, and 26.7% [24.3,29.4] of Gini change. These are within-observable fractions and are not a common-scale comparison. Summing adjacent dissimilarities gives 2.458, 5.327, and 1.525, while the corresponding endpoint dissimilarities are .453, .763, and .321. This is an operational sum over saved checkpoints, not a claim that all three quantities define geometric distances.

Held-out behavior audit. The behavior manifest freezes the first prospective targeted-intervention split's 128 prompts (disjoint from the Stage-2 analysis set and zero-overlap with the SFT corpus), greedy decoding, 512 new tokens, seed 20261006, and last-number correctness parser for every state. The completed results are:

Table 17: Held-out behavior along the controlled trajectory. Cap is the fraction of generations reaching 512 new tokens.

Step	Correct/128	Accuracy	Mean tokens	Cap
0	85	.664	59.1	.008
50	75	.586	151.2	.164
100	73	.570	96.4	.102
150	72	.562	158.1	.172
200	84	.656	308.5	.359
250	69	.539	341.8	.453
300	79	.617	328.1	.398
350	66	.516	366.5	.523
400	79	.617	335.3	.398
450	65	.508	374.2	.547
500	66	.516	375.1	.555
600	63	.492	368.1	.547
700	68	.531	393.7	.594
800	62	.484	406.8	.672
900	68	.531	408.9	.688
1000	62	.484	426.3	.719

Every state contains 128 unique records and a passing summary. Because later outputs are frequently right-censored at the generation cap, accuracy and length should be read jointly.

Reorganization–vulnerability pilot audit. The baseline phase evaluates 1,536 candidate sites (64 traces, one stratified generation position, and 24 downstream layers), then freezes 54 interventions with six in each $O^B \times \mathcal{R}$ cell. Reorganization cut points are .571 and .750; log-risk cut points are -2.064 and -1.506. Selection and outcome SHA256 values begin `0abeb788ec5e116e` and `679609fbb464bad8`. All selection-hash, active-source, baseline-repeat, target-route, finite-value, and confirmation-isolation checks pass. The 54 sites span 40 prefixes; prefix reuse across different cells is retained in the cluster bootstrap.

Common-Trace score replay audit. The score replay stores 36,853 prompt-plus-continuation tokens for each of Base, A3B, and Thinking, with 48 layers, top-16 IDs, and top-16 FP32 router scores. Each score stream passes record-count, uniqueness, shape, descending-score, normalized-weight, content-hash, and confirmation-isolation checks. The score record SHA256 values begin `7fff8f83b37db70b`, `860f492e19ec66b0`, and `589a29b9d0004042`; the analysis summary is stored under `artifacts/graph/common_trace_score_decomposition_n64/`. The permutation-control summary is stored under `artifacts/graph/expert_permutation_control_qwen_n64/`. Its Hungarian map is fit separately for every layer on 32 alternating prefixes and evaluated on the other 32, then the folds are reversed. Fingerprint channels are early/middle/late top-16 score, top-16 presence, top-8 activation, and selected gate weight. All reported Jaccard changes use held-out prefixes and 20,000 prefix-bootstrap draws. The route-optimized control is stored under `artifacts/graph/route_optimized_permutation_qwen_n64/`. It uses the same folds and learns each layer's assignment from binary top-8 activation vectors over 16,126 or 16,218 training-context token sites; no modified-forward outcome enters either assignment. For Thinking–Base, the two folds improve the mean training activation-Jaccard objective by .00047 and .00052 over identity; their non-fixed fractions are .123 and .121.

Full SHA256 values are stored in the released manifests. For compact audit, their leading 16 hexadecimal characters are: first prospective-study prefix/selection/outcome `e45109de55c77ecf`, `287d825c913ed2ff`, and `be8aa423eee30525`; second prospective-study prefix/selection/outcome `114a664f7a1131e4`, `490d7ba687b3e948`, and `650783637ac75daf`. A3B selection/outcome begin `bb615e3a3f6b8ab5`/`2478bcbdde1836e2`; Thinking begins `1dc57f6e46fdb8f5`/`6377ee7825d3be1f`. Corrected-score Nemotron Base selection/outcome begin `7b72886c99438242`/`c3bfe808238a4ba9`; Nemotron Post begins `e1fc1b80a63d2626`/`fb89b604a1164e44`. All baseline repeats, counts, target boundaries, finite values, and selection hash checks pass. No confirmation data are used.

Component and non-max prospective-control freeze. The new control prefix, component-selection, component-outcome, non-max-selection, and non-max-outcome SHA256 values begin `58969e9184dc3819`, `f80c92e39385f701`, `54b9ad4bff0117a5`, `7d87e7a7763aff6d`, and `e0a44cef3471313f`. The two 64-prefix subsets are disjoint from both earlier prospective targeted-intervention splits. Both outcome programs verify the frozen selection hash before loading the model and never recompute selection features.

Table 18: Prospective non-max ordering conditional on near-boundary candidates. One downstream event is sampled per prefix before outcomes; all eight active source directions are retained. Intervals are prefix-cluster bootstrap 95% CIs.

Risk quintile	n	Frozen $\mathcal{R}$ range	Exact target flip
Q1	103	$[2.0\times10^{-6}, 7.8\times10^{-4}]$	.000 [.000,.000]
Q2	103	$[8.2\times10^{-4}, 2.9\times10^{-3}]$	.000 [.000,.000]
Q3	102	$[2.9\times10^{-3}, 8.1\times10^{-3}]$	.000 [.000,.000]
Q4	102	$[8.1\times10^{-3}, 4.0\times10^{-2}]$	.000 [.000,.000]
Q5	102	$[4.0\times10^{-2}, 3.20]$	.118 [.024,.219]

Table 19: Cross-checkpoint selected-event geometry and local susceptibility.

Checkpoint	Median γ	Median $\lvert g\rvert$	Median $\lvert g\rvert/\gamma$	Random flip	High/low
Base	0.00229	0.0787	30.44	0.1172	4.54×
A3B	0.00174	0.0664	29.00	0.1250	4.31×
Thinking-2507	0.00305	0.0547	15.91	0.0703	5.00×

For 128 paired Base–Thinking prefixes, selected-event γ is larger for Thinking in 73 cases (57.0%) and smaller in 55; the paired median ratio is 1.44. The median derivative over all eight active sources is smaller in 72 cases (56.2%) and larger in 56; its paired median ratio is 0.83. Hence the aggregate

margin shift is statistically resolved but heterogeneous, not pointwise dominance. The stored basis contains all eight source directions only at the selected target event; it cannot recover the unconditional distribution over every nonselected eligible event.

Common-site boundary turnover. Freezing the Base-selected physical (ℓ, t) does not freeze the underlying decision boundary. Qwen Thinking preserves the Base rank-boundary expert pair in only 1/128 prefixes, and Kanana Thinking in only 2/128. At those physical sites, the state-specific Thinking/Base geometric-mean ratios are 40.5 [27.4,60.2] and 52.0 [35.9,75.4] for γ, 0.34 [0.28,0.41] and 0.29 [0.23,0.38] for maximum active-source $|g|$, and 0.0083 [0.0053,0.0129] and 0.0056 [0.0036,0.0087] for maximum $|g|/\gamma$. Because the boundary expert pair changes in nearly every prefix, these are state-specific physical-site operating-point contrasts, not same-boundary geometric effects.

Table 20: Qwen source-layer utilization and functional identity across the checkpoint progression. Utilization is restricted to physical layer 23 on the 512-prefix build set; source-match rates use the 128-prefix prospective split. These utilization values have a different layer, input, and aggregation scope from both the controlled-trajectory and all-layer atlas summaries.

State	Entropy	N_{eff}	Top-16 Jaccard vs. Base	High-risk source match
Base	3.455	31.7	1.000	1.000
A3B	3.490	32.8	.882	.266
Thinking	3.445	31.3	.684	.281

Nemotron amendment and superseded path. Nemotron routes by $r_i = \sigma(q_i) + b_i$, not by raw logits. We therefore define its rank-6/rank-7 margin in corrected score space. An initial raw-logit-boundary analysis is retained only as a superseded diagnostic and is not reported as a scientific result. The corrected protocol independently freezes all 128 target events before modified forwards, perturbs only routed experts (not the shared expert), and uses a central finite-difference derivative with step 10^{-3}. Both corrected 384-record outcome grids pass selection-hash, uniqueness, finite-value, exact final-logit repeat, and exact target-boundary checks. Maximum absolute selection-to-outcome margin drift is 3.81×10^{-6} for Base and 8.39×10^{-5} for Post, while boundary expert identities remain exact.

Nemotron high-risk flip rates over ascending within-model $\mathcal{R}$ quintiles are $[0, 0, 0.038, 0.32, 0.84]$ for Base and $[0, 0, 0, 0.28, 0.84]$ for Post. These bins are diagnostic within-model calibration, not a shared probability scale. Their sharp ordering complements the cross-family functional summary: the risk primitive transfers even though the Post/Base effect and geometry contrasts are unresolved. A multi-step FD audit at eight frozen events fails for both Nemotron states: relative to 10^{-3}, median relative L2 errors are 1.00 at 3×10^{-4} and 0.66–0.71 at 3×10^{-3}, with low sign agreement. The discrete derivatives are FP32 cancellation-limited. This negative numerical result does not alter the already frozen 10^{-3} prospective enrichment, but it rules out strong cross-step claims about Nemotron's derivative magnitude or checkpoint shift.

Kanana architecture and derivative-associated variation. Kanana-2 uses a DeepSeek-V3-style MLA/MoE implementation with 48 decoder layers: layer 0 is dense and layers 1–47 are MoE. Each MoE layer has 128 routed experts, top-6 selection, and one shared MLP whose width equals two routed experts. We use physical source layer 23 and destinations 24–47, intervene only on routed contributions, and leave the shared MLP untouched. Expert identities are selected by $\sigma(q) + b$; contribution weights instead gather uncorrected sigmoid scores, renormalize over top-6, and multiply by 2.448. This explicitly separates ranking score from contribution weight. Stock/out-of-place aggregation relative L2 is 6.18×10^{-8} for Base and 6.04×10^{-8} for Thinking.

Base and Thinking selection/outcome hashes begin `e3d6e497867da070`/`df86a9ca7703c59e` and `3951599cfa5e1f25`/`6da1b7ab6dc96a71`. Each checkpoint uses a distinct 512-prefix atlas and a pre-outcome frozen selection. Both 384-record grids pass exact inactive/zero-delta, baseline repeat, boundary, finite-value, uniqueness, and selection-hash checks, with no relaxed matches. Ascending-risk flip rates are $[0, .038, 1, 1, .92]$ and $[0, 0, .231, .92, .96]$. The nonmonotonic top-bin fluctuation is compatible with 25–26 prefixes per bin; the defensible claim is sharp low-to-high stratification, not deterministic calibration.

Both checkpoints use the identical plain prompt rather than the Thinking chat template. Consequently the comparison is cleaner with respect to

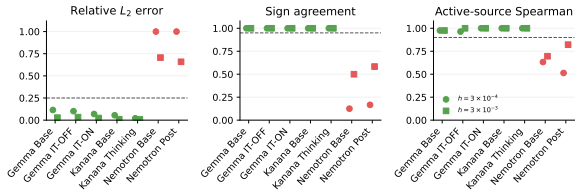


Figure 19: Multi-step finite-difference audit relative to the frozen $h = 10^{-3}$ oracle. Gemma and Kanana conditions pass the preregistered relative error, sign, and rank gates at 3×10^{-4} and 3×10^{-3}; both Nemotron states fail. Green/red indicate the overall condition-level decision; dashed lines are acceptance thresholds.

Table 21: Multi-step derivative audit. Each pair reports the median at $h = 3 \times 10^{-4}$ / 3×10^{-3} relative to the frozen 10^{-3} estimate. Spearman and sign agreement are over active source directions.

Condition	Relative L_2	Sign agreement	Spearman	Decision
Gemma Base	.115/.028	1.00/1.00	.976/.976	Pass
Gemma IT-OFF	.102/.034	1.00/1.00	.964/1.00	Pass
Gemma IT-ON	.070/.024	1.00/1.00	1.00/1.00	Pass
Kanana Base	.055/.010	1.00/1.00	1.00/1.00	Pass
Kanana Thinking	.021/.008	1.00/1.00	1.00/1.00	Pass
Nemotron Base	1.00/.706	.125/.500	.632/.696	Fail
Nemotron Post	1.00/.659	.167/.583	.514/.822	Fail

input context, but remains an observational weight-state association and does not evaluate generated reasoning. A 12-event follow-up varies the central-FD step over 3×10^{-4}, 10^{-3}, and 3×10^{-3}. Relative to 10^{-3}, median relative L2 errors are 0.055/0.010 for Base and 0.021/0.008 for Thinking; sign agreement and active-source Spearman are 1.0 throughout. The derivative stability gate therefore passes, closing the step-size concern while leaving the selected-event comparison limitation intact.

Gemma architecture and mode-conditioned context. The executable Gemma 4 26B-A4B model has 30 MoE text layers, 128 routed experts, top-8 selection, and no shared expert. Its router first computes FP32 softmax probabilities, selects top-8 identities, then renormalizes selected weights and applies per-expert scales. We therefore define γ as the rank-8/rank-9 gap in the pre-selection softmax probabilities. Stock in-place and derivative-safe out-of-place aggregation differ by relative L2 7.09×10^{-8}. All three 512-prefix atlases and 384-record outcome grids pass inactive, zero-delta, repeat, boundary, finite-value, and selection-hash checks; no matched-low selection is relaxed.

Eight-event multi-step FD audits also pass for Base, IT-OFF, and IT-ON. Relative to the frozen 10^{-3} step, median relative L2 errors at 3×10^{-4} are 0.115, 0.102, and 0.070, and at 3×10^{-3} are 0.028, 0.034, and 0.024; sign agreement is 1.0 throughout and median active-source Spearman is at least 0.964. Thus all reported Gemma conditions pass the same numerical stability gate.

Gemma Base, IT-OFF, and IT-ON high-risk flip rates over ascending $\mathcal{R}$ quintiles are $[0, 0, 0, 0, 0.76]$, $[0, 0, 0.115, 0.92, 0.96]$, and $[0, 0, 0, 0.36, 0.96]$. OFF and ON use identical IT weights but distinct official chat renderings. In an audited prefix they contain 68 and 71 tokens and share no terminal suffix: OFF appends an empty thought block, whereas ON prepends a system turn with <|think|>. Accordingly, the resolved ON/OFF geometry shift is evidence for mode-conditioned context dependence, not an isolated causal effect of thinking.